

A-level

ENVIRONMENTAL SCIENCE

Paper 2 - Generated practice paper Set 36

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	• $(240\,000 - 2\,000) / 240\,000 \times 100 = 99.2\%$	1
01.2	• Whales are k-selected: slow growing, late maturing, one calf every few years, so populations recover very slowly • Technology (harpoon cannons, steam / factory ships) allowed catches far above the sustainable yield within decades • No regulation / open access to the ocean; whales were valuable (oil, spermaceti) and easily found	3
01.3	• Moratorium on commercial whaling since 1986 until populations recover • Total protection for certain species; whale sanctuaries • Limits on numbers and sizes taken; protection of suckling mothers and calves • Research into whale biology and threats (ship strikes, net entanglement); exceptions: aboriginal subsistence quotas (eg Inuit bowheads), 'scientific' permits (Japan - ruled not research in 2014), Norway / Iceland set their own quotas	4
01.4	• ITTO's 1990 target of trade only from sustainably managed forests by 2000 was not met and little progress since • Timber comes from many countries / private land with weak enforcement, high demand and alternative land uses; whaling involves few nations and a single treaty body	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 31-32; AQA spec 3.1.2.3.2.	
02.1	• Negative feedback responds to any change by re-establishing the previous equilibrium • Only large unusual events overwhelm the feedback so a new equilibrium forms • Example: methane hydrate release at the end of the Permian causing climate change and mass extinction	3

Question	Marking guidance	Marks
02.2	• In a diverse system each component is relatively less important so the loss / change of one has a small effect; alternative food sources and relationships buffer the change • Natural: coral reefs and tropical rainforests with complex food webs are stable; simple polar / desert systems fluctuate • Human: agroecosystems focus on a few crops and remove detritivores, pollinators and predators so are vulnerable to pests, disease, weather • Diverse technical systems: a society using a range of renewable energy resources is more secure than one relying on a single source	4
02.3	<i>Any 3 from:</i> • Melting permafrost releasing methane • Reduced albedo as ice / snow melt • Methane hydrate release • Forest and peat fires • Faster soil decomposition • Ocean acidification reducing sequestration • Cirrus cloud formation	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 371-372; AQA spec 3.6.1.	
03.1	• Light: selective felling to create clearings for plants needing high light while retaining dense canopy for shade species; clearings also give warm sand / soil for lizard eggs; leave dead wood; woodland floor plants (bluebells) grow before the canopy leaves • Dissolved oxygen: narrowing a slow river increases flow and turbulence so more oxygen dissolves, suiting trout and mayfly larvae; shallow water warms quickly for egg development • Nutrients: removing fertile topsoil / not fertilising allows plants of nutrient-poor soils (eg sundew on acidic nitrate-deficient soils) that cannot compete on fertile ground; dung patches add local nutrients	6
03.2	<i>Any 4 from:</i> • Food: wildflower grasslands for seed-eating birds; providing food species • Control of predation: island nest sites unreachable by foxes; trapping / culling / eradicating introduced rats (Scilly Isles - Manx shearwater and storm petrel breeding success increased) • Control of competitors: removing rhododendron so wildflowers get light • Pollination: a range of flowering plants with different flowering times so pollinators have food all season • Seed dispersal species (hippos, elephants, monkeys) conserved; control of introduced pathogens	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 43-46; AQA spec 3.1.2.3.4, 3.1.3.1.	
04.1	• Its reading is adjusted to match a meter known to be accurate so that the results are accurate / comparable • If it cannot be adjusted the percentage error should be recorded so the results can be corrected	2

Question	Marking guidance	Marks
04.2	<i>Any 3 from:</i> • Droppings: diet, gender, territory size • Owl pellets: diet • Feeding marks (eg chewed hazel nuts): presence of species such as dormice • Nests / burrows: presence and breeding • Tracks / footprints: presence, movements • Territorial marks (scratching posts): presence and territory	3
04.3	• Bat detectors / sonograms record the ultrasonic sounds bats produce, which differ between species • The recordings provide evidence of presence, abundance and activity of species that cannot easily be seen	2
04.4	<i>Any 3 from:</i> • Marked individuals mix freely with the rest of the population • Marks do not fall off or wear away, and marking does not affect survival • The population does not change during the study (no births, deaths or migration) • All individuals have an equal chance of being caught / animals are mobile and can be caught	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 402, 405, 407; AQA spec 3.1.2.3.5-3.1.2.3.6.	
05.1	• $2.7 - 2.45 = 0.25$ billion = 250 million years	1
05.2	• For millions of years all the oxygen produced reacted with (dissolved) iron in the oceans • Only when all the iron had reacted did surplus dissolved oxygen build up in the oceans and then escape into the atmosphere	2
05.3	<i>Any 3 from:</i> • Oxygen in the atmosphere absorbed ultraviolet light • A dynamic equilibrium of reactions between O, O₂ and O₃ (the ozone layer) developed • Less UV reached the surface so organisms were no longer restricted to water (which had absorbed the UV) • Aerobic respiration became possible so more complex / larger organisms (plants and animals) evolved; many anaerobic Archaea and bacteria died out	3
05.4	• Photoautotrophs absorbed carbon dioxide, a greenhouse gas, and some of the carbon was stored in geological sediments such as carbonate rocks and fossil fuels (carbon sequestration) • The lower CO₂ concentration reduced the retention of heat, offsetting the increase in the brightness of the Sun (about 10% per billion years)	2
05.5	<i>Any 2 from:</i> • Biogeochemical cycles developed so small amounts of nutrient elements could support life for long periods without depletion • Transpiration by land plants returned water vapour to the atmosphere and increased rainfall in other areas • Regulation of atmospheric composition (eg oxygen / CO₂ by photosynthesis and respiration)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 6-8; AQA spec 3.1.1.2.	

Question	Marking guidance	Marks
06.1	- Natural processes are driven by renewable, low energy-density solar energy captured by photosynthesis; human processes usually use high energy-density non-renewable fossil fuels - Natural processes occur at low temperatures because enzymes lower the activation energy of reactions; human processes need high temperatures and pressures - Natural processes produce no toxic wastes / their products are raw materials for other processes; industrial processes emit CO₂ and NO_x - So natural fixation is sustainable indefinitely with a low carbon footprint	4
06.2	<i>Any 3 from:</i> - Hydrological cycle (evaporation, precipitation) - Carbon cycle / photosynthesis - Atmospheric circulation / winds - Thermohaline circulation (density differences from heating / evaporation) - Decomposition	3
06.3	- Bioleaching and phytomining extract metals at low temperature instead of high-temperature smelting - Anaerobic digestion and composting treat wastes; enzymes in industry (biological washing powders, biofuels from algae / microorganisms) - Heat pumps and solar thermal capture low-density heat; low-temperature manufacturing using biological polymers; fuel cells instead of combustion	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 373; AQA spec 3.6.2.	
07.1	- Spearman's rank correlation coefficient - Mann-Whitney U test - Chi-squared test - t-test	4
07.2	- Chi-squared compares frequencies / counts in categories - Dissolved oxygen is continuously variable measured data, so a t-test or Mann-Whitney U test is needed	2
07.3	- There is always some remaining uncertainty / a probability that the result was produced by chance - Statistics only give a level of confidence (eg 95%, 99%) that the result is significant	2
07.4	- Bars drawn above and below each mean showing plus and minus one standard deviation - If the bars of two means overlap a lot the difference is probably not significant; if they do not overlap it probably is	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 422-423; AQA spec 3.7.2.3.	
08.1	- The range of an environmental factor within which a species / population can survive (survival highest at the optimum, falling towards the limits) - X survives from about pH 5.5 to 8.5 (wide range); Y only from about pH 6.0 to 8.0 (narrow range) - Both have the same optimum (pH 7.0)	3
08.2	- X: 60% survive at pH 6.0 whereas only 5% of Y survive - X has a larger gene pool / wider range of tolerance so some individuals are adapted to the new conditions - Survivors breed and their offspring are adapted so the population rebuilds; Y is likely to die out locally	3

Question	Marking guidance	Marks
08.3	• Reveals the habitat requirements / which abiotic factor is near the edge of tolerance • Management can make small beneficial changes to that factor (eg liming, shading, water levels) to keep conditions within the range and increase survival	2
08.4	<i>Any 2 from:</i> • Food supply • Pollinators • Seed dispersal species • Disease / pathogens • Predators / competitors	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 72-73; AQA spec 3.1.3.1.	
09.1	<i>Any 2 from:</i> • Hedgerows • Hay meadows • Grazed moorland / heathland • Chalk grassland	2
09.2	• Drainage produces more aerobic soils so wetland plants and the animals they support cannot survive • Fertilisers increase growth of tall species such as grasses which out-compete smaller plants such as wildflowers	2
09.3	• The introduced (non-indigenous) predator may eat other species as well as the pest	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 328-329; AQA spec 3.5.1.3.	

Question	Marking guidance	Marks
10.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Context: wild catches have not increased since about 1990 as stocks are fully or over-exploited; aquaculture output growing rapidly and approaching the wild catch; fish are the last large-scale hunted food • Advantages of aquaculture: control of limiting factors, selective breeding, gender control, high yields per area; fish have low FCRs (no thermoregulation / support) compared with livestock; herbivores (tilapia, carp) and molluscs very efficient; polyculture, IMTA, aquaponics, rice-fish • Limits: carnivorous species (salmon, shrimp) need fishmeal from wild fish so still depend on fishing and deplete forage fish (sandeels - puffins); some species (bluefin tuna) will not breed in captivity; consumer preference for carnivores • Environmental impacts: habitat loss (mangroves), organic pollution, pesticides, antibiotics and resistance, lice, escapes and gene pool contamination, predator control; and how they can be reduced • Fishing can be sustainable with quotas, gear design, NTZs, size limits, effort limits; MSY management needs good data • Judgement: aquaculture can supply much of the growth in demand, especially herbivores and shellfish, but cannot fully replace fishing; both need sustainable management	9
10.2	• A species that is given artificial food (pellets) rather than relying on natural food in the water, eg salmon, shrimp	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 338-359; AQA spec 3.5.2.	

Question	Marking guidance	Marks
11.1	Discuss how an understanding of energy flow and the manipulation of food species can be used to increase food production sustainably. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Energy flow: autotrophs and heterotrophs; losses by reflection, transmission, respiration; 100 000 to 1 000 units; few food chains beyond four levels; plant food most productive; grazing animals where crops impossible; pigs on wastes • Energetics: intensive vs extensive, law of diminishing returns, energy subsidies (fertiliser, pesticides, irrigation, machinery, drying, processing, transport), energy ratios, food conversion ratios, dependence on fossil fuels • Control of genetics: asexual reproduction / cuttings / cloning, selective breeding and breeds, cross-breeding and hybrid vigour, AI and embryo transfer • GM: Roundup Ready soya, Bt crops, golden rice, omega-3 oilseed rape; advantages; risks (gene transfer, antibiotic markers, allergies, costs to LEDC farmers, loss of local diversity) • Sustainability links: energy dependence, trophic level of diet, feeding 12 billion people • Justified conclusion on the balance between productivity, efficiency and sustainability Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 321-328; AQA spec 3.5.1.2.	25

Question	Marking guidance	Marks
11.2	Discuss the extent to which agriculture can be made sustainable while feeding a growing human population. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Need: population possibly 12 billion; humans raise carrying capacity; trophic level of diet • Impacts to be reduced: habitat loss, drainage, nutrient enrichment, introduced species, pesticide and nutrient pollution, nitrate toxicity, greenhouse gases, hydrological changes, soil erosion • Social, economic and political influences: consumer choices, technology availability, grants, guaranteed markets, surpluses, quotas, set-aside, agri-environment schemes • Strategies: cultural and biological pest control, integrated control, organic nutrient supply, legumes, low tillage, water conservation, CWR conservation, renewable energy, methane reduction, habitat retention • Limits: little new land, imports reducing local availability, cost and acceptance of new techniques, past land degradation • Justified conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 328-337; AQA spec 3.5.1.3-3.5.1.5.	25